

Chirath Hettiarachchi

Applied ML | Computational Neuroscience | Biomedical AI Systems | Reinforcement Learning

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SUMMARY

Research Scientist and ML Engineer with a PhD in Computer Science and experience developing reinforcement learning, biomedical AI, computational neuroscience, and health informatics systems. Lead ML Researcher on translational healthcare projects involving closed-loop clinical treatment, simulation-based evaluation, and collaboration with clinicians, engineers, researchers, industry partners, and lived-experience contributors. Built open-source research software, reproducible experimental pipelines, and user-facing clinical AI platforms, with prior industry experience developing enterprise ML systems for financial technology applications.

EDUCATION & RESEARCH HIGHLIGHTS

Doctor of Philosophy, Computer Science — RL-based Artificial Pancreas Systems Sep 2019 – Jun 2023
Australian National University (ANU) *Australia*

- Thesis: An RL-based artificial pancreas system to fully automate treatment in type 1 diabetes.
- ★ **Scientific Impact:** Developed G2P2C, a modular RL algorithm and demonstrated in U.S. FDA-accepted preclinical simulations, that RL-based insulin delivery can eliminate manual meal announcements and carbohydrate estimation, thus reducing patient burden [Paper].
- ★ **Engineering Impact:** Maintained open-source research software tools for researchers (RL4H ☆71 ⭐33) and developed a GPU-based high-performance simulations (GluCoEnv) to accelerate diabetes research [Code].
- ★ **Research Translation Impact:** Built and maintained CAPSML, a deployed clinical AI web platform used by 2,600+ users across 67 countries, supporting human-centred AI development, interactive education, analysis, clinical and lived-experience stakeholder feedback, and research translation.
- ★ **Awards:** Outstanding New Researcher Highly Commended Award, ACT Health Directorate, 2025; HEX Innovation Startup Program Scholarship 2023 (ANU); ANU PhD Scholarship and HDR Merit Scholarship.

M.Sc. (Research), Computer Science and Engineering Nov 2018 – Aug 2020
University of Moratuwa *Sri Lanka*

- Thesis: ML for type 2 diabetes prediction from photoplethysmography & physiological data.
- ★ **Awards:** Prof. V.K.Samaranayake Research Grant (LK Domain Registry, 2019); Student Travel Grant (AIME'19).

B.Sc. Eng. (Hons.), Electronic and Telecommunication Engineering Jan 2014 – Jan 2018
University of Moratuwa *Sri Lanka*

- Thesis: A wearable system for analysing human arm motion and predicting throwing-related injury risk.

CIMA Passed Finalist (Professional Qualification) Aug 2012 – Nov 2014
Chartered Institute of Management Accountants (CIMA) *United Kingdom*

- ★ **Awards:** Sri Lankan Prize Winner, Performance Management (2013).

EXPERIENCE

Research Fellow — RL-based Brain Stimulation Oct 2023 – Present
Australian National University, School of Computing *Canberra, Australia*

- Lead ML Researcher on an MRFF-funded translational project developing home-based brain stimulation treatment for depression. Coordinated ML research across multidisciplinary stakeholders including computer scientists, clinicians, engineers, industry partners, and lived-experience contributors. Supervised and mentored 1 PhD, 1 Master's, and 9 Honours/research students, translating complex ML and biomedical research problems into feasible projects, codebases, documentation, and publishable outputs.
- Developed computational simulations and learning methods for ML-based adaptive clinical treatment, with emphasis on reinforcement learning, computational neuroscience, control systems, and biomedical signal analysis.
- Designed and maintained HPC/cloud workflows on NCI Australia's Gadi supercomputer and Nirin Cloud for large-scale simulation, model evaluation, and reproducible biomedical AI experiments.
- ★ **Scientific Impact:** Demonstrated preliminary simulation evidence that RL-guided stimulation can shift depression-associated neural circuit dynamics toward healthier patterns [Paper].
- ★ **Engineering Impact:** Built an open-source simulation framework for evaluating reinforcement learning strategies for non-invasive brain stimulation in depression, connecting computational neuroscience models, stimulation protocols, closed-loop control, and RL algorithms [Code: NeuroStimEnv].
- ★ **Awards:** Commendation for Excellence in Supervision, ANU, 2025; SMP ECR Research Impact Award (Diagnostics, Therapeutics, and Clinical Care), ANU, 2024; ANU ECR Travel Grant, 2026.

Honorary Fellow – Diabetes Technology Research

St Vincent's Hospital, University of Melbourne

Jun 2021 – Jun 2024

Melbourne, Australia

- Collaborated with clinical and engineering researchers on next-generation multi-input artificial pancreas systems.
- Applied ML, reinforcement learning, glucose prediction, and biomedical signal processing to physiological data for decision-support and automated insulin delivery research.

Research Officer – Wearable Sensor and Digital Health Software

Australian National University, School of Computing

Sep 2022 – Sep 2023

Canberra, Australia

- Developed software for digital health research using web, mobile, and smartwatch data sources including Apple Watch (watchOS), Samsung Galaxy (Wear OS), and Fitbit devices.
- Built ML analysis and visualisation workflows for extracting and interpreting physiological signals from wearables.
- ★ **Awards:** Branko Cesnick Best Academic Paper Award, HIC 2024.

Machine Learning Engineer

Yaala Labs

Dec 2018 – Sep 2019

Sri Lanka

- Developed enterprise-level ML systems for financial technology applications including name screening and financial risk prediction for global financial markets.

Data Analytics and Machine Learning Engineer

Apr 2018 – Sep 2018

Research Intern

Forestpin

Aug 2016 – Dec 2016

Sri Lanka

- Analysed organisational data across diverse industries and conducted research to develop machine learning models for better analysis and forecasting.
- Developed statistical and ML algorithms for the Forestpin Risk Engine product, a forensic tool to identify outliers and discrepancies in corporate financial transactions.

Co-Founder / CEO – Digital Health Startup

Olivescript

Jan 2017 – Sep 2018

Sri Lanka

- Co-founded and led a digital health startup developing a healthcare device and predictive analytics platform for family health monitoring. Led early product development, stakeholder engagement, and ML translation.
- ★ **Awards:** Commonwealth Digital Health Awards (Winner, Non-Communicable Disease Category, 2017); Disrupt Asia Startup Battle (Winner, 2017); and Microsoft Imagine Cup Sri Lanka (2nd Runner-Up, 2017).

TECHNICAL SKILLS

Core Expertise: Applied machine learning, reinforcement learning, deep learning, generative modelling, time-series modelling, biomedical signal processing, anomaly detection, model evaluation

Translational & Human-Centred AI: Digital health, clinical AI translation, human-centred AI, closed-loop clinical decision systems, multidisciplinary stakeholder engagement, clinician/consumer-informed research

Research Engineering: Python, PyTorch, scikit-learn, MATLAB, Simulink, MLflow, Git

Scientific Simulation: NEURON, LFPy, SimNIBS, UVA/PADOVA Type 1 Diabetes Simulator, computational neuroscience simulations, physiological simulations, clinical treatment simulation

Data Modalities: Electroencephalography (EEG), photoplethysmography (PPG), electromyography (EMG), inertial measurement unit (IMU) sensor, smartwatch/mobile data streams, blood biomarkers, and financial transaction data

Software & Data Systems: Web/mobile/smartwatch data pipelines, Nginx, research data processing workflows

Infrastructure: NCI Australia HPC (Gadi supercomputer, Nirin Cloud), DUG Australia, AWS, Microsoft Azure, reproducible experiment workflows, open-source research software

COURSES, TRAINING, & CERTIFICATIONS

- **Elevate Deep-Tech Startup Workshop**, CICADA Innovations, 2025.
- **Good Clinical Practice (GCP), in Australia** (Western Australian Health Translation Network)(2025-present, 2022-2024).
- **Understanding AS ISO/IEC 42001:2023, Information Technology - Artificial Intelligence - Management System**, Standards Australia, 2024.
- **Generative AI with Large Language Models**, DeepLearning.AI, Coursera.
- **ACADI Mentoring Program**, Australian Centre for Accelerating Diabetes Innovations, 2026; **NECTAR Mentoring Program**, Network for Early-Career Teachers, Academics, and Researchers, ANU, 2024.
- **Work Health and Safety Risk Management Training; Responding to Disclosures of Sexual Violence Training; Emergency Control Organisation (Warden) Training; Equal Opportunity Training**, ANU.

REFERENCES

Available upon request.

October 2026